

Justin Tawan Ward

TIGA Graduate of the Year - Programmer

Projects

February 2025 - January 2026 | Freelance Industry Brief - XR Group Project

Blackbeard's Cannon Clash | Unity, C#, VR, Git

- A standalone VR experience for installation in the National Maritime Museum's Pirate Exhibition, following a brief they delivered, meetings with staff at the museum, and updates over the period of the exhibition
- Set sail on Blackbeard's ship and hear about his stories from the NPC version of Blackbeard
- Look out for enemy ships with his spyglass and fire his cannons to defeat them

Mechanics Programmer and VR Optimisation

- I set up the VR interactions aboard the ship and created the spyglass and cannon game mechanics
- I used optimisation techniques to allow the game to run at a smooth framerate as a standalone Android build for Quest devices
- Fixing bugs throughout the development process and with each update from the museum

September 2024 - April 2025 | Final Year Individual Project

Echoes Act | Unity, C#, Git, ShaderGraph, HLSL

- Performing research into Non-Human First Person Perspectives in games
- Specifically looking into echolocation and the effect that a new sense has on immersion
- Play as a dolphin who uses echolocation to traverse the dark environment and clean up ocean rubbish

Solo Game Developer

- I used Unity's ShaderGraph to create intersection shaders for the echolocation pulses and HLSL to outline objects which the player can interact with
- Uses 3D audio and haptic feedback to increase immersion in low-visibility situations
- Performed quantitative and qualitative data gathering to determine the level of immersion which the echolocation mechanics provide for the player

September 2024 - December 2024 | Final Year Group Project

Gone Fishin' | Unity, C#, VR, Git

TIGA Best Student Game 2025 Finalist

- Relaxing Virtual Reality fishing game with graphics inspired by SEGA Bass Fishing
- Catch 6 types of fish, with normally distributed weights, which affect catch difficulty
- Following an Agile approach to project management

Lead Programmer

- I programmed the fishing rod's mechanics, using physics-based casting and reeling for realism
- I set up the VR interactions with the fish, rod, and other objects in the scenes using Unity's XR Interaction Toolkit

Septemter 2023 - April 2024 | Solo Project

Temporus | Unreal Engine 5, Blueprints, C++

- A 3D Hack'n'Slash platformer
- Advanced character controller - slide, dash, wall run, edge mantle
- Time slowing mechanics

Solo Game Developer

- I developed the game using Unreal Engine' blueprint system, creating the player and gameplay loop
- Created enemy AI using UE5's blackboard and behaviour trees
- Created interactable pick up items using custom C++ classes

November 2023 - January 2024 | Group Project

So You're Tellin' Me? | Unity, C#, Arduino, C++, Git

TIGA Best Student Game 2024 Finalist

- 3D cooking game with unique installation
- A custom controller shaped like a pan uses an Arduino to send data to Unity
- Uses Arduino input components like a gyroscope, potentiometer, joystick, and outputs like LEDs, motor.

Lead Programmer

- I programmed the game in Unity and coded the ability for information to be passed between the Arduino and Unity
- I created the main mechanics and gameplay loop, including the player's input affecting the wok in game and the progression system, as well as outputs to the Arduino components
- Showed my teamworking and communication skills while the 3 of us worked remotely over the holidays

Contact

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Education

University of Greenwich

BSc (Hons) Games Design and Development

First Class Honours

School Prize for Outstanding Undergraduate Achievement

John Taylor High School

A Level:

- Computer Science - A
- Maths - A
- Physics - B

Skills

- C#
- Unity
- C++
- Unreal Engine 5
- JavaScript
- Version Control
- OOP
- Game Development

Work Experience

○ Freelance Game Programmer - Present

National Maritime Museum

Hybrid - Greenwich/Remote

- I worked as the mechanics and VR optimisation programmer on a team of 3, creating a VR game for the Pirate's Exhibiton at the National Maritime Museum in 2025
- I worked to the given breif from the museum, with updates and progression on the game throughout the development cycle and period of the exhibition

○ Unity Tutor - 1 Year

Spark4Kids

Remote

- I tutor Unity and C# to DofE aged children (aged 13-17) following Spark4Kid's Unity course
- Acting as a supervisor during the sessions, offering help to any student who has questions about Unity or C# code

○ Cafe Assistant - 7 Months

The Galley Cafe

Greenwich, London

- Serving food, working on the tills, cleaning tables and surfaces, and working as a barista
- Handling stressful situations with lunchtime rushes and ensuring swift service for customers

○ Kitchen Porter - 14 Months

The Meynell Ingram Arms

Staffordshire

- Ensuring cleanliness throughout the kitchen throughout the shift in the kitchen and in customer's dishes and cutlery.
- Assisting chefs and other staff with their duties, occasionally making pizzas

○ Race Marshal - 18 Months

Midland Karting

Staffordshire

- Teaching safety measures to customers and how to safely operate the karts
- Marshalling the track to look out for dangers or help customers
- Working resiliently in many different weather conditions and across long shifts

Achievements

School of Computing and Mathematical Sciences Prize for Outstanding Undergraduate Achievement

At my university graduation ceremony, when receiving my degree, I also received the School Prize for Outstanding Undergraduate Achievement in Games Design and Development for graduating with the highest grades on the course.

TIGA UK Games Education Awards

2024 - My group project "[So You're Tellin' Me...?](#)" was nominated for the Best Student Game 2024. We were shortlisted for the finals against 12 other projects from university games courses around the UK. This was a great milestone as a second year student at the time, and the team are all so proud of reaching the finals!

2025 - I was shortlisted as a finalist for Outstanding TIGA Graduate of the Year: Programmer for my First-Class grades, projects, and showreel, where I received the accolade of TIGA Graduate of the Year - Programmer for reaching the finals. My final year group project "Gone Fishin'" also reached the finals in the Best Student Game 2025 category, competing against 9 other student games.

Digital Shark Expo - UoG End of Year Show

2023 - I was nominated to present "[Dauntless Depths](#)" as a first year, which was a great acheivement as the show was used to show off final year work in its previous years.

2024 - I showcased my second year group project "[So You're Tellin' Me...?](#)" to great enjoyment of the audience and players, and solo UE5 project "[Temporus](#)", which was mistaken for a commercial game by the Head of School.

2025 - I presented 5 of my final year projects: "[Echoes Act](#)", "[Gone Fishin'](#)", my [Shader Programming portfolio](#), "[Blackbeard's Cannon Clash](#)", and "[Murhpy's Gift Delivery Service](#)", showing my skills of confidence and communication when speaking to industry professionals about my games, and potential new students about the course, as well as my multi-tasking skills throughout my final year to produce 5 high quality projects.

UOG Games Dev Society

I was a founding member of the new Games Dev Society at the University of Greenwich during my second year at uni. My main role was Instagram Manager, but I also showed my leadership and management through my involvment in the planning and running of events, as well as advertising and getting new people interested. We won the Best New Academic Society Award at the Student Led Teaching Awards 2024!

DofE Gold Award

I received my Duke of Edinburgh Gold Award in Buckingham Palace Gardens and spoke with the Duke of Edinburgh, Prince Edward. I completed a week residential at Scorpion Airsoft where I learnt good teamwork skills with a new group of people. I also completed a 6-day expedition in Mid Wales, walking over 100km and climbing to the peak of Pumlumon Fawr - with a 752m elevation. I used my skills to communicate with my group, boost morale and keep everyone motivated, ending with my group being the first group to finish.